

GNOmE: Graph Networks for Mechanistic Explicability

Sehaj Singh
Independent Research
sehajr-singhs@github.com

Abstract—We present GNOmE (Graph Networks for Mechanistic Explicability), a framework that extracts computation graphs from trained neural networks and uses graph neural networks to predict unit importance without intervention experiments. GNOmE treats each attention head and MLP layer as a graph node, with edges weighted by Jacobian-based contribution strengths. A GCN reader trained on these graphs achieves $r = 0.475$ Pearson correlation in predicting head importance across independently trained models, demonstrating that computational graph structure encodes transferable functional information.

Index Terms—mechanistic interpretability, graph neural networks, transformer circuits, Jacobian analysis

I. INTRODUCTION

Mechanistic interpretability aims to understand *how* neural networks compute, not just what they compute. Circuit analysis [1], [2] identifies subgraphs responsible for specific behaviors, but existing methods like path patching [3] scale quadratically with model size.

We propose extracting the computation graph directly from model weights and activations, then using a GNN to predict functional properties. This is $O(N \cdot L)$ versus $O(N^2)$ for patching, and enables cross-model transfer.

II. METHOD

Graph Extraction. For a transformer with L layers and H heads, we create nodes for each head L_kH_h and MLP layer. Edges connect consecutive layers with weights:

$$w_{uv} = \left| \frac{\text{mean}_{b,t}[\mathbf{c}_u] \cdot \text{mean}_{b,t}[\mathbf{c}_v]}{\|\text{mean}_{b,t}[\mathbf{c}_u]\| \cdot \|\text{mean}_{b,t}[\mathbf{c}_v]\|} \right| \quad (1)$$

where $\mathbf{c}_u$ is unit u 's contribution to the residual stream.

GNN Reader. A 2-layer GCN predicts per-node importance from the graph:

$$\hat{\mathbf{y}} = \text{ReLU}(\hat{\mathbf{A}} \cdot \text{ReLU}(\hat{\mathbf{A}}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)\mathbf{W}_3 \quad (2)$$

III. EXPERIMENTS

Setup. We train 2-layer, 4-head transformers ($\sim 460\text{K}$ params) on the induction task [2]: predict B given $[A][B][\dots][A]$. Three independently seeded models achieve $>95\%$ accuracy.

Results. Table I shows key metrics.

Cross-Model Transfer. Leave-one-out GNN evaluation yields $r = 0.456, 0.520, 0.449$ (mean **0.475**), showing graph structure predicts function across model instances.

TABLE I
EXPERIMENTAL RESULTS

Metric	Model 0	Model 1	Model 2
Accuracy	0.950	0.967	0.951
Edges	25	25	25
Attr.–Patching r	0.508	0.529	0.509

IV. CONCLUSION

GNOmE extracts computation graphs from trained models and reads explicability from them via GNNs, achieving meaningful cross-model transfer without intervention experiments. Future work will scale to larger models and evaluate on diverse circuit types.

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